

Exploring Human-Generative AI Collaboration in Software Testing



Mandatory questions are marked with a star (*)

This survey is part of a Master's Thesis at the University of Oulu, exploring collaboration between professionals and Generative AI in software testing. Your input will help advance understanding of this emerging area in practice.

Please review the Informed Consent provided below before starting. If you have any questions, you can contact the researcher at chamari.dasanayake@student.oulu.fi.

Thank you for your time and contribution.

1. Informed Consent *

- ☐ I confirm that I have read and understood the [Informed Consent for Participation in Research](#) and voluntarily agree to participate in this research study.

Section 1: Background

This section collects basic information about your role and experience in software testing. It helps us understand the diversity of professional backgrounds among participants.

2. What is your current (or most recent) role related to software testing or quality assurance? (Select the closest match.) *

- ☐ Test / QA Engineer
- ☐ Test / QA Lead or Manager
- ☐ Test Automation Engineer
- ☐ Software Engineer (involved in testing)
- ☐ DevOps / Release Engineer (involved in testing)
- ☐ Other (please specify) _____

3. How many years of experience do you have in software testing or QA? *

- ☐ Less than 1 year
- ☐ 1-3 years
- ☐ 4-6 years
- ☐ 7-10 years
- ☐ 11-15 years
- ☐ More than 15 years

4. Which of the following types of testing do you regularly engage in? (Select all that apply) *

- ☐ Unit Testing
- ☐ Integration Testing
- ☐ Functional Testing
- ☐ System Testing
- ☐ End-to-End Testing
- ☐ Performance / Load Testing
- ☐ Security Testing
- ☐ UI/UX Testing
- ☐ Other (please specify) _____

5. In which business domain or industry does your company primarily operate? (Select the closest match) *

- ☐ Finance / Banking
- ☐ Healthcare / Medical
- ☐ Telecommunications
- ☐ Retail / E-commerce
- ☐ Manufacturing
- ☐ Government / Public Sector
- ☐ IT / Software Services
- ☐ Education

☐ Other (please specify) _____

6. Approximately how many people are involved in software development (including testing) in your organization? *

- ☐ Fewer than 10
- ☐ 10–49
- ☐ 50–99
- ☐ 100–249
- ☐ 250–499
- ☐ 500 or more

7. In which country are you currently based? *

- ☐ Afghanistan
- ☐ Albania
- ☐ Algeria
- ☐ Andorra
- ☐ Angola
- ☐ Antigua and Barbuda
- ☐ Argentina
- ☐ Armenia
- ☐ Australia
- ☐ Austria
- ☐ Azerbaijan
- ☐ Bahamas
- ☐ Bahrain
- ☐ Bangladesh
- ☐ Barbados
- ☐ Belarus
- ☐ Belgium
- ☐ Belize
- ☐ Benin

- ☐ Bhutan
- ☐ Bolivia
- ☐ Bosnia and Herzegovina
- ☐ Botswana
- ☐ Brazil
- ☐ Brunei
- ☐ Bulgaria
- ☐ Burkina Faso
- ☐ Burundi
- ☐ Cabo Verde
- ☐ Cambodia
- ☐ Cameroon
- ☐ Canada
- ☐ Central African Republic
- ☐ Chad
- ☐ Chile
- ☐ China
- ☐ Colombia
- ☐ Comoros
- ☐ Congo (Brazzaville)
- ☐ Congo (Kinshasa)
- ☐ Costa Rica
- ☐ Croatia
- ☐ Cuba
- ☐ Cyprus
- ☐ Czech Republic
- ☐ Denmark
- ☐ Djibouti
- ☐ Dominica
- ☐ Dominican Republic
- ☐ Ecuador
- ☐ Egypt

- ☐ El Salvador
- ☐ Equatorial Guinea
- ☐ Eritrea
- ☐ Estonia
- ☐ Eswatini
- ☐ Ethiopia
- ☐ Fiji
- ☐ Finland
- ☐ France
- ☐ Gabon
- ☐ Gambia
- ☐ Georgia
- ☐ Germany
- ☐ Ghana
- ☐ Greece
- ☐ Grenada
- ☐ Guatemala
- ☐ Guinea
- ☐ Guinea-Bissau
- ☐ Guyana
- ☐ Haiti
- ☐ Honduras
- ☐ Hungary
- ☐ Iceland
- ☐ India
- ☐ Indonesia
- ☐ Iran
- ☐ Iraq
- ☐ Ireland
- ☐ Israel
- ☐ Italy
- ☐ Ivory Coast

- ☐ Jamaica
- ☐ Japan
- ☐ Jordan
- ☐ Kazakhstan
- ☐ Kenya
- ☐ Kiribati
- ☐ Kuwait
- ☐ Kyrgyzstan
- ☐ Laos
- ☐ Latvia
- ☐ Lebanon
- ☐ Lesotho
- ☐ Liberia
- ☐ Libya
- ☐ Liechtenstein
- ☐ Lithuania
- ☐ Luxembourg
- ☐ Madagascar
- ☐ Malawi
- ☐ Malaysia
- ☐ Maldives
- ☐ Mali
- ☐ Malta
- ☐ Marshall Islands
- ☐ Mauritania
- ☐ Mauritius
- ☐ Mexico
- ☐ Micronesia
- ☐ Moldova
- ☐ Monaco
- ☐ Mongolia
- ☐ Montenegro

- ☐ Morocco
- ☐ Mozambique
- ☐ Myanmar
- ☐ Namibia
- ☐ Nauru
- ☐ Nepal
- ☐ Netherlands
- ☐ New Zealand
- ☐ Nicaragua
- ☐ Niger
- ☐ Nigeria
- ☐ North Korea
- ☐ North Macedonia
- ☐ Norway
- ☐ Oman
- ☐ Pakistan
- ☐ Palau
- ☐ Palestine
- ☐ Panama
- ☐ Papua New Guinea
- ☐ Paraguay
- ☐ Peru
- ☐ Philippines
- ☐ Poland
- ☐ Portugal
- ☐ Qatar
- ☐ Romania
- ☐ Russia
- ☐ Rwanda
- ☐ Saint Kitts and Nevis
- ☐ Saint Lucia
- ☐ Saint Vincent and the Grenadines

- ☐ Samoa
- ☐ San Marino
- ☐ Sao Tome and Principe
- ☐ Saudi Arabia
- ☐ Senegal
- ☐ Serbia
- ☐ Seychelles
- ☐ Sierra Leone
- ☐ Singapore
- ☐ Slovakia
- ☐ Slovenia
- ☐ Solomon Islands
- ☐ Somalia
- ☐ South Africa
- ☐ South Korea
- ☐ South Sudan
- ☐ Spain
- ☐ Sri Lanka
- ☐ Sudan
- ☐ Suriname
- ☐ Sweden
- ☐ Switzerland
- ☐ Syria
- ☐ Taiwan
- ☐ Tajikistan
- ☐ Tanzania
- ☐ Thailand
- ☐ Timor-Leste
- ☐ Togo
- ☐ Tonga
- ☐ Trinidad and Tobago
- ☐ Tunisia

- ☐ Turkey
- ☐ Turkmenistan
- ☐ Tuvalu
- ☐ Uganda
- ☐ Ukraine
- ☐ United Arab Emirates
- ☐ United Kingdom
- ☐ United States
- ☐ Uruguay
- ☐ Uzbekistan
- ☐ Vanuatu
- ☐ Vatican City
- ☐ Venezuela
- ☐ Vietnam
- ☐ Yemen
- ☐ Zambia
- ☐ Zimbabwe

Section 2: Context

This section explores your familiarity with and usage of Generative AI tools in the context of software testing.

8. How familiar are you with Generative AI tools? *

- ☐ Not at all familiar
- ☐ Slightly familiar
- ☐ Moderately familiar
- ☐ Very familiar
- ☐ Extremely familiar

9. Have you personally used any Generative AI tools as part of your software testing work? *

- ☐ Yes, I use Generative AI regularly in my testing work
- ☐ Yes, I have experimented with Generative AI a few times
- ☐ Not yet, but I am interested in trying Generative AI in testing
- ☐ No, and I am not interested in using Generative AI for testing

10. If you have used Generative AI in your testing work, please briefly describe one example of how you used it or what you used it for. (If you have not used Generative AI in testing, you may skip this question.)

Section 3: Benefits

This section focuses on your perceptions of the potential benefits of using Generative AI in software testing tasks.

11. Please indicate how much you agree or disagree with each of the following statements about the potential benefits of using Generative AI in software testing. *

[illegible]

	Strongly Agree	Agree	Somewhat Agree	Somewhat Disagree	Disagree	Strongly Disagree	I don't know / No opinion
Generative AI can increase test coverage by generating a wider range of test cases. *	☐	☐	☐	☐	☐	☐	☐
Generative AI helps with debugging by identifying issues or providing insights into code. *	☐	☐	☐	☐	☐	☐	☐
Generative AI improves test quality by reducing human error and providing consistent test results. *	☐	☐	☐	☐	☐	☐	☐
Generative AI helps optimize testing *	☐	☐	☐	☐	☐	☐	☐
Collaboration between human testers and AI leads to better testing outcomes than either alone. *	☐	☐	☐	☐	☐	☐	☐

12. Please indicate any other benefits of using Generative AI in testing.

Section 4: Challenges

This section asks about the challenges or difficulties you may have encountered (or anticipate) when using Generative AI in testing.

13. Please indicate your level of agreement with each of the following statements about challenges in using Generative AI for software testing. *

[illegible]

14. Please indicate any other challenges of using Generative AI in software testing.

Section 5: Opportunities

This section explores what future applications or improvements in Generative AI for testing you find most interesting or valuable.

15. Which of the following potential applications of Generative AI in software testing do you find most promising or exciting? (Select up to 3) *

- ☐ Automated generation of test cases or test scripts from requirements or code.
- ☐ AI-assisted testing of graphical user interfaces (GUI).
- ☐ AI-assisted testing of end-to-end (E2E) system scenarios.
- ☐ Hybrid human-AI collaborative workflows.
- ☐ Applying AI in early-stage testing such as in requirements or design phase.
- ☐ Integrating generative AI with traditional testing tools and frameworks.
- ☐ Using AI for non-functional (quality) testing.
- ☐ Access to more mature, ready-to-use AI-powered testing tools that can be deployed in real projects.

16. Please indicate any other promising opportunities of using Generative AI in software testing.

Section 6: Risks

This section focuses on your concerns or perceived risks related to the use of Generative AI in software testing.

17. Which of the following risks or concerns about using Generative AI in software testing do you consider most critical? (Select up to 3) *

- ☐ Incorrect results or assertions causing false positives/negatives in tests
- ☐ Irrelevant or inaccurate AI-generated content for the testing task
- ☐ Context window (token limits) causing missing or truncated information, leading to incomplete analysis or suggestions
- ☐ Lack of transparency making it hard to trust or explain the outcomes
- ☐ Privacy and security risks due to exposure of sensitive code or data
- ☐ Biases or blind spots introduced in test cases
- ☐ Difficulty in scaling or maintaining AI-generated test assets over time

18. Please indicate any other risks or concerns of using Generative AI in software testing.

Section 7: Final Thoughts

This final section provides space for any additional comments, suggestions, or experiences you'd like to share related to human-AI collaboration in software testing.

19. Do you have any other comments, suggestions, or experiences regarding human-AI collaboration in software testing that you would like to share?

20. If you would like to receive a summary of the survey results, you may enter your email address. (Your email will be used only to share the results and will not be linked to your survey responses.)

Email
